Data-Driven Analysis of Groundwater Trends and Stress Zones in India

1. Project Overview:

- Groundwater is an essential water source in India, widely used for agriculture, domestic, and industrial purposes. However, excessive usage and changing environmental conditions have led to declining groundwater levels in many regions.
- This project, titled **"Data-Driven Analysis of Groundwater Trends and Stress Zones in India"**, analyzes real data collected by the "Central Ground Water Board (CGWB)". The study focuses on identifying groundwater trends, seasonal variations, and stress zones across different regions. The findings help in understanding groundwater depletion patterns and support sustainable water management practices.

Project Objective

- To analyze groundwater
- level changes in different regions of India.
- To identify areas experiencing serious groundwater depletion.
- To study seasonal variations in water levels.
- To classify regions based on groundwater stress levels.
- To visualize trends using charts and maps.
- To create awareness about the importance of groundwater conservation.

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2. Data Sources

- **Source Description:**

  This dataset is maintained by the Central Ground Water Board (CGWB) and provides detailed measurements of changes in groundwater levels from a network of 22,965 observation wells across India. The data, which has been collected four times annually (January, March/April/May, August, and November) since 1969, specifically records the fluctuations in the depth to groundwater over time, offering crucial insights into both seasonal and annual variations.

- **Source Link:**

https://indiadataportal.com/p/groundwater/r/mojs-wris_cgwb_wells_level_changes-pl-ot-aaa

- **Timeline:** The dataset was created and last updated in 2025 (approximately), and contains groundwater level data from 2013 to 2023."

- **Domain:** Environmental Science with Data Analytics

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3. Problem Statement

- To analyze groundwater levels across different states, districts, and basins to find areas facing water shortage and areas with good availability.
- To study year-wise changes in groundwater levels to understand long-term trends and patterns.
- To compare current water levels and level differences to identify regions with serious groundwater decline.
- To classify regions into safe, moderate, critical, and over-exploited zones for better understanding of groundwater conditions.
- To create new measures such as stress index to evaluate groundwater risk using available data.
- To present groundwater trends through charts and maps for easy interpretation and decision-making.
- To understand regional differences in groundwater availability to support sustainable water management.

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4. Original Dataset Attributes (Column /Features) Details:

Attribute Name	Description	Data Type
Id	Unique identification number for each record	Number
Date	Date of observation	Date
state_name	Name of the state	Text
state_code	Unique code of the state	Number
district_name	Name of the district	Text
district_code	Unique code of the district	Number
station_name	Name of the observation station	Text
latitude	Latitude of the station	Number (With decimal points up to 5)
longitude	Longitude of the station	Number (With decimal points up to 5)
basin	Name of the river basin	Text
sub_basin	Name of the sub-basin	Text

source	Data provider (CGWB)	Text
Current level	Current groundwater level (meters below ground level)	Number (With decimal points up to 2)
level_diff	Change in groundwater level from previous period	Number (With decimal points up to 2)

4.1 Final Model Attributes (Column /Features) Details:

After data modelling, the original dataset was normalized into fact and dimension tables. Descriptive attributes were shifted to dimension tables, and only keys and measures were retained in the fact table.

Attribute Name	Description	Data Type
ID	Unique identification number for each record	Number
Date	Date of observation	Date
Station-Code	Unique code of the Station	Text
Current_Depth_to_Water_Level	Current groundwater level (meters below ground level)	Number (With decimal points up to 2)
Level_Difference	Change in groundwater level from previous period	Number (With decimal points up to 2)
Depth_Category	Groups groundwater measurements into meaningful levels to make analysis easier.	Text
Trend_Type	Used to classify groundwater observations into Decline and Recovery based on Level_Difference values	Text

5. Tools & Technologies

- **Microsoft Office Professional Plus 2019(64-Bit):** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

- **Data Cleaning & Transformation:**
  - The dataset contains records from 2013 to 2023; however, data for 2022 and 2023 was available only for a limited number of states. Therefore, to ensure consistency and completeness across regions and to manage dataset size, the analysis was restricted to the period from 2019 to 2021.
  - The original dataset contains approximately 5 lakh records. After filtering, the dataset size was reduced to approximately 1,25,000 rows.
  - Sub-basin-wise analyses were excluded to maintain clarity and focus on broader groundwater trends at the state, district, and basin levels.
  - Removed the **Source** column from the main dataset since all records originated from the Central Ground Water Board (CGWB).
  - Assigned appropriate data types to all columns in the dataset.
  - Applied TRIM, CLEAN, and PROPER functions to text columns to remove unwanted spaces and standardize text formatting.

```
=PROPER(TRIM(CLEAN([@[State_Name]])))
```

```
=PROPER(TRIM(CLEAN([@[District_Name]])))
```

```
=PROPER(TRIM(CLEAN([@[Station_Name]])))
```

```
=PROPER(TRIM(CLEAN([@Basin])))
```

  - To follow the official naming format, standardized the state names for **Dadra and Nagar Haveli and Daman and Diu**, **Jammu and Kashmir**, and **Andaman and Nicobar Islands**.
  - Corrected district names to match official spellings and formats (e.g., Angul, Kushinagar, East Singhbhum, Morigaon).
  - Standardized outdated, incorrect, and alternate names to their official versions (e.g., Anugul → Angul, Korea → Koriya, Marigaon → Morigaon).
  - Updated district names where subdivisions or informal names were used (e.g., South Andamans → South Andaman).
  - Created a separate correction sheet to store verified standard station names.
  - Standardized the station names by removing numeric suffixes using below formula (e.g., Station1 → Station) where ever it is applicable.

```
=IF(AND(ISNUMBER(RIGHT(C2,1)*1),COUNTIFS(
$A$2:$A$17690,A2,
$B$2:$B$17690,B2,
$D$2:$D$17690,D2,
$E$2:$E$17690,E2,
$C$2:$C$17690,LEFT(C2,LEN(C2)-1)
)>0),LEFT(C2,LEN(C2)-1),C2)
```

- Corrected identified misspellings using a validated reference list.
- Applied INDEX and MATCH functions to map corrected names to the main dataset.

```
=INDEX(Sheet8!$F$2:$F$17690,
MATCH([@[Station_Name]],Sheet8!$C$2:$C$17690,0))
```

- Generated a new column named "Standard Station Name" to store the cleaned and standardized values.
- Ensured original station names were retained when no verified correction was available.
- Created a **Depth-Category** column in the main dataset based on **Current_Depth_to_Water_Level** values using below formula.

```
=IF([@[Current_Depth_to_Water_Level]]<=5,"Shallow",IF([@[Current_Depth_to_Water_Level]]<=15,"Medium","Deep"))
```

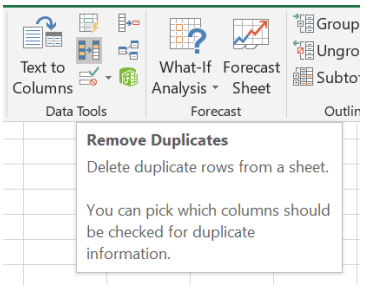
- ## Converting Data into Fact and Dimensional Tables:
  - After completing data cleaning (handling missing values, correcting spelling errors in State/District names, standardizing formats, and creating calculated fields), a Hybrid **Snowflake Schema** was designed for efficient storage, scalability, and hierarchical analysis.
  - The model consists of:
    - **1 Fact Table**
    - **Multiple Dimension Tables**
    - **Two Hierarchies**
      - Administrative Hierarchy
      - Hydrological Hierarchy

- ## Schema Type: Hybrid Snowflake Schema
  - The model follows a **hybrid snowflake** structure, where geographical dimensions are normalized, while the Date dimension is partially denormalized for performance optimization. This **Schema** was chosen instead of a star schema because:
  - Administrative data has hierarchical levels (Region → State → District → Station)
  - Hydrological grouping (Basin → Station) forms another hierarchy
  - Avoids redundancy in State and Basin data
  - Improves normalization and data integrity
  - Although Snowflake Schema is used for location dimensions, the Date dimension is kept denormalized by including month and year attributes in a single table to improve performance and simplify analysis in Excel.

- **Administrative Hierarchy:**
  - **Structure:** Region→ State → District → Station → Fact Table
  - **Dim_Region Table:** A separate **Region Dimension (Dim_Region)** was created to standardize regional classification, where each region is assigned a unique region code (R01–R06) to ensure consistency, reduce redundancy, and support region-wise analysis of groundwater data.

Region	Region_Code
North	R01
South	R02
East	R03
West	R04
Central	R05
North-East	R06

  - **Dim_State Table:** Developed a State Dimension table containing State_Name, State_Code and Region_Code attributes for structured geographical analysis.
    - Extracted and retained distinct state names and corresponding state codes using Excel's Advanced Filter functionality.



    - For each state, the corresponding region code is mapped using an Excel formula to ensure accurate and consistent regional classification.

```
=INDEX(Dim_Region!$B$2:$B$7,MATCH([@Region],Dim_Region!$A$2:$A$7,0))
```

- Screenshot of Dim_State Table:

State_Name	State_Code	Region-Code
Jammu and Kashmir	1	R01
Himachal Pradesh	2	R01
Punjab	3	R01
Chandigarh	4	R01
Uttarakhand	5	R01
Haryana	6	R01
Delhi	7	R01
Rajasthan	8	R01
Uttar Pradesh	9	R01
Bihar	10	R03
Arunachal Pradesh	12	R06
Nagaland	13	R06
Tripura	16	R06
Meghalaya	17	R06
Assam	18	R06
West Bengal	19	R03
Jharkhand	20	R03
Odisha	21	R03
Chhattisgarh	22	R05
Madhya Pradesh	23	R05
Gujarat	24	R04
Maharashtra	27	R04
Andhra Pradesh	28	R02
Karnataka	29	R02
Goa	30	R04
Kerala	32	R02
Tamil Nadu	33	R02
Puducherry	34	R02
Andaman and Nicobar Islands	35	R03
Telangana	36	R02
Dadra and Nagar Haveli and Daman and Diu	38	R04

- **Dim_District Table:** Created a District Dimension table to support district-level geographical analysis.
  - Extracted the following columns from the raw dataset:
    - **State_Code**
    - **District_Name**
    - **District_Code**
  - Removed repeated district records and retained only unique combinations of State_Code and District_Code.
  - Verified duplicate records using the following COUNTIFS formula, which returned **"1" for all rows**, confirming that no duplicates were present.

```
=COUNTIFS($A$2:$A$559,A2,$B$2:$B$559,B2,$C$2:$C$559,C2)
```

  - Structured the table to establish proper hierarchical relationships with the State table.
  - Reference Screenshot of Dim_District Tablefor first 20 rows shown below:

State_Code	District_Name	District_Code
36	Adilabad	501
9	Agra	118
24	Ahmedabad	438
27	Ahmednagar	466
8	Ajmer	86
27	Akola	467
32	Alappuzha	554
9	Aligarh	119
23	Alirajpur	639
8	Alwar	87
6	Ambala	58
9	Ambedkar Nagar	121
27	Amravati	468
24	Amreli	439
3	Amritsar	27
9	Amroha	154
24	Anand	440
28	Ananthapuramu	502
21	Angul	344

Dim_Station Table:

- The Station Dimension Table (Dim_Station) is a core component of the Snowflake Schema used in this project. It stores all master information related to groundwater monitoring stations and acts as a bridge between administrative and basin hierarchies and the fact table.
- This table ensures data consistency, reduces duplication, and enables efficient analysis.
- To associate basin information with state data, a Basin_Code column was first created in the main dataset using the INDEX and MATCH functions, referencing the Basin dimension table where basin codes were maintained.

```
=INDEX(Dim_Basin!$B$2:$B$24,MATCH('Fact GroundWater'!J2,Dim_Basin!$A$2:$A$24,0))
```

- After mapping the basin codes, only the required attributes were extracted from the main dataset to construct the Station dimension table.
- District_Code and Basin_Code were added to the Station dimension to link each station with its corresponding administrative unit and hydrological basin, enabling hierarchical analysis across both structures.
- The Dim_Station table is created from the main groundwater dataset using the following steps:
    - Selected relevant columns from the main dataset.
    - Removed duplicate station records using Advanced Filter
    - Standardized station names
    - Generated unique Station Codes using below excel formula

```
="ST" & TEXT(ROW([@[Station_name]])-1,"000")
```

- Validated latitude and longitude values

- Reference Screenshot of first 20 rows of Dim_Station Table:

District_Code	Basin-Code	Station_Name	Standard_Station_Name	Latitude	Longitude	Station_Code
632	B06	Laxmipur	Laxmipur	13.28556	93.00306	ST001
632	B06	Durgapur	Durgapur	13.23083	92.95528	ST002
632	B06	Joypur	Joypur	12.69917	92.88167	ST003
632	B06	Subhasgram	Subhasgram	13.24083	92.99056	ST004
632	B06	Nabagram	Nabagram	13.17333	92.93194	ST005
632	B06	Shibpur	Shibpur	13.24278	93.04417	ST006
632	B06	Tikadera	Tikadera	12.45667	92.77361	ST007
632	B06	Mayabunder	Mayabunder	12.87667	92.88111	ST008
632	B06	Nimbutala	Nimbutala	12.52472	92.97028	ST009
632	B06	Betapur	Betapur	12.62222	92.87278	ST010
632	B06	Padmanavapuram	Padmanavapuram	12.50833	92.96056	ST011
632	B06	Basantipur	Basantipur	12.72917	92.88472	ST012
632	B06	Uttara	Uttara	12.31611	92.78889	ST013
632	B06	Haribeck	Haribeck	12.98222	92.84556	ST014
632	B06	Kadamtala	Kadamtala	12.34472	92.77889	ST015
632	B06	Kausalyanagar	Kausalyanagar	12.39528	92.78222	ST016
632	B06	Pahalgaon	Pahalgaon	12.83944	92.85333	ST017
632	B06	Tugapur No 8	Tugapur No 8	12.83972	92.85389	ST018
632	B06	Phooltala Basti	Phooltala Basti	12.41111	92.76222	ST019

Dim_Basin Table:

- Extracted all Basin names from the main dataset and copied them into a separate worksheet for preprocessing.
- Used Advanced Filter in Excel to remove duplicate Basin names and retain only unique values.
- Created a new column named Basin-Code to assign a unique identifier to each Basin.
- Generated Basin codes using the formula:

```
="B" & TEXT(ROW(A2)-1,"00")
```

- Screenshot of Dim_Basin Table:

Basin	Basin-Code
Area Of Inland Drainage In Rajasthan	B01
Barak and Others	B02
Brahamaputra	B03
Brahmani and Baitarni	B04
Cauvery	B05
Drainage Area Of Andaman and Nicobar Islands Basin	B06
East Flowing Rivers Between Mahanadi and Pennar	B07
East Flowing Rivers Between Pennar and Kanyakumari	B08
Ganga	B09
Godavari	B10
Indus (Up To Border)	B11
Krishna	B12
Mahanadi	B13
Mahi	B14
Minor Rivers Draining Into Myanmar and Bangladesh	B15
Narmada	B16
Pennar	B17
Sabarmati	B18
Subernarekha	B19
Tapi	B20
West Flowing Rivers From Tadri To Kanyakumari	B21
West Flowing Rivers From Tapi To Tadri	B22
West Flowing Rivers Of Kutch and Saurashtra Including Luni	B23

- **Dim_Date Table:**
  - Created a Date Dimension table containing Date, Year, Month, and Month_Name attributes to support time-based analysis.
  - **Date:** Generated continuous dates from January 1, 2019, to December 31, 2021, using below Excel formula.
  -
```
=IF((DATE(2019,1,1)+ROW(A2)-2)<=DATE(2021,12,31),DATE(2019,1,1)+ROW(A2)-2,"")
```
  - **Year:** Created a Year column using below Excel-based formula for annual trend analysis.

```
=YEAR(A2)
```

  - **Month:** Derived a Month column using an Excel formula to support monthly-level analysis.

```
=MONTH([@Date])
```

  - **Month_Name**: Generated a Month_Name column using an Excel formula to improve data readability and reporting.

```
=TEXT(A2,"MMM")
```

  - **Monsoon Season:** A column named Monsoon_Season in the main dataset created based on the **period** when **Depth_to_Water_Level** was measured.

```
=IF(AND(MONTH([@date])>=6,MONTH([@date]<=9)), "Post-Monsoon","Pre-Monsoon")
```

  - Reference Screenshot of **Date-Dim** table containing 20 rows (Original Date-Dim table has 1096 rows)

Date	Year	Month	Month_Name	Monsoon-Season
01-01-2019	2019	1	Jan	Pre-Monsoon
02-01-2019	2019	1	Jan	Pre-Monsoon
03-01-2019	2019	1	Jan	Pre-Monsoon
04-01-2019	2019	1	Jan	Pre-Monsoon
05-01-2019	2019	1	Jan	Pre-Monsoon
06-01-2019	2019	1	Jan	Pre-Monsoon
07-01-2019	2019	1	Jan	Pre-Monsoon
08-01-2019	2019	1	Jan	Pre-Monsoon
09-01-2019	2019	1	Jan	Pre-Monsoon
10-01-2019	2019	1	Jan	Pre-Monsoon
11-01-2019	2019	1	Jan	Pre-Monsoon
12-01-2019	2019	1	Jan	Pre-Monsoon
13-01-2019	2019	1	Jan	Pre-Monsoon
14-01-2019	2019	1	Jan	Pre-Monsoon
15-01-2019	2019	1	Jan	Pre-Monsoon
16-01-2019	2019	1	Jan	Pre-Monsoon
17-01-2019	2019	1	Jan	Pre-Monsoon
18-01-2019	2019	1	Jan	Pre-Monsoon
19-01-2019	2019	1	Jan	Pre-Monsoon
20-01-2019	2019	1	Jan	Pre-Monsoon

- **Fact_Ground_Water:**
  - Since a snowflake schema was implemented, the fact table should directly connect to Dim_Station and Dim_Date tables through Station_Code and Date respectively.
  - Therefore, all other administrative and geographical codes were removed from the fact table and maintained in their corresponding dimension tables

to preserve normalization and avoid redundancy.

- A separate column named **Station-Code** was added to Fact table using the below excel formula.

```
=INDEX('Station-Dim'!$H$2:$H$17690,MATCH([@[Standard_Station_Name]],'Station-Dim'!$E$2:$E$17690,0))
```

- The Station_Code was included in the fact table as a foreign key to establish a direct relationship between each groundwater measurement and its corresponding monitoring station.
- This enables accurate linking of transactional data with station-related attributes stored in the Station dimension and supports location-based analysis.
- The "**Depth_Category"** column was derived from the current ground-to-water level measurements using a conditional Excel formula, classifying observations into **Shallow, Medium, and Deep** categories for analytical purposes.

```
=IF([@[Current_Depth_to_Water_Level]]<=5,"Shallow",IF([@[Current_Depth_to_Water_Level]]<=15,"Medium","Deep"))
```

- A new derived column named "**Trend_Type"** was created based on the **Level_Difference** attribute to classify groundwater observations into Decline and Recovery categories.
- Records with negative level difference values were labeled as "Recovery", indicating groundwater depletion, while positive values were labeled as "Decline", indicating recharge.
- This classification enabled frequency-based analysis of groundwater trends across states, districts, and regions.
- Each record in the fact table represents one groundwater observation for a specific station on a given date.
- Reference screenshot of first 20 rows of Fact_Ground_Water table as shown below:

ID	Date	Station_Code	Current_Depth_to_Water_Level	Level_Difference	Depth_Catagory	Trend_Typ
11	04-04-2019	ST001	0.12	-0.33	Shallow	Recovery
12	29-11-2019	ST001	1.47	1.35	Shallow	Decine
23	04-04-2019	ST002	0.49	-0.19	Shallow	Recovery
24	29-11-2019	ST002	1.31	0.82	Shallow	Decine
34	04-04-2019	ST003	0.88	-0.07	Shallow	Recovery
35	29-11-2019	ST003	0.97	0.09	Shallow	Decine
47	04-04-2019	ST004	0.15	-0.23	Shallow	Recovery
48	29-11-2019	ST004	1.43	1.28	Shallow	Decine
60	29-11-2019	ST005	1.00	0.48	Shallow	Decine
72	04-04-2019	ST006	0.24	-0.41	Shallow	Recovery
73	29-11-2019	ST006	1.38	1.14	Shallow	Decine
84	04-04-2019	ST007	2.17	-0.10	Shallow	Recovery
85	29-11-2019	ST007	2.56	0.39	Shallow	Decine
90	04-04-2019	ST008	0.47	-0.25	Shallow	Recovery
91	29-11-2019	ST008	1.62	1.15	Shallow	Decine
101	04-04-2019	ST009	0.47	-0.57	Shallow	Recovery
102	29-11-2019	ST009	1.81	1.34	Shallow	Decine
111	04-04-2019	ST010	0.18	-0.57	Shallow	Recovery
112	29-11-2019	ST010	1.27	1.09	Shallow	Decine

- ## Data Analysis Using Pivot Tables:
  - ### State-wise Average Groundwater Depth:

State_Name	Avg of Current_Depth_to_Water_Leve
Andaman and Nicobar Islands	3.51
Andhra Pradesh	5.72
Arunachal Pradesh	4.58
Assam	4.42
Bihar	4.40
Chandigarh	25.03
Chhattisgarh	5.13
Dadra and Nagar Haveli and Daman and Diu	4.36
Delhi	14.91
Goa	4.89
Gujarat	12.21
Haryana	22.49
Himachal Pradesh	5.63
Jammu and Kashmir	4.67
Jharkhand	4.87
Karnataka	6.68
Kerala	6.30
Madhya Pradesh	6.96
Maharashtra	6.44
Meghalaya	2.67
Nagaland	3.62
Odisha	3.18
Puducherry	11.63
Punjab	16.32
Rajasthan	21.27
Tamil Nadu	9.94
Telangana	8.27
Tripura	7.32
Uttar Pradesh	9.79
Uttarakhand	12.43
West Bengal	5.96

(Higher depth = deeper water table = higher stress)

- **Insights:**
  - States showing **very high average depth (>20 m)** experiencing critically deep groundwater levels, indicating heavy extraction and significant water stress.
  - States which have avg Depth from 10m to 20 m are under moderate to high stress and may face worsening conditions if extraction continues.
  - States which have moderate groundwater depth(6-10m), suggesting relatively balanced recharge and usage.
  - States which have low Stress / shallow water level (<6 m) indicating healthy recharge and lower stress conditions.
  - There is clear **regional imbalance** in groundwater availability across India. Urbanized and agriculturally intensive states show deeper water levels, while high rainfall regions maintain shallow groundwater levels.

  - ### Region-wise Groundwater Depth Analysis

Regions	Avg of Current_Depth_to_Water_Level
Central	6.28
East	4.36
North	14.65
North-East	4.55
South	7.94
West	8.47

- **Insights:**
  - North region has the highest average groundwater depth, while East and North-East have the lowest.
  - Higher depth in the North is likely due to intensive agriculture, urban demand, and over-extraction, whereas high rainfall supports recharge in the East and North-East.
  - If current extraction continues, groundwater levels in North and West regions are likely to decline further in the coming years.
  - Priority should be given to implementing recharge structures and regulating groundwater usage in high-stress regions, especially in North and West India.

- **Basin-wise Groundwater Depth Analysis**

Basin_Name	Avg of Current_Depth_to_Water_Level
Area Of Inland Drainage In Rajasthan	29.05
Barak and Others	5.31
Brahmani and Baitarani	3.57
Brahmaputra	4.23
Cauvery	6.08
Drainage Area Of Andaman and Nicobar Islands Basin	3.51
East Flowing Rivers Between Mahanadi and Pennar	3.99
East Flowing Rivers Between Pennar and Kanyakumari	11.80
Ganga	8.23
Godavari	6.25
Indus (Up To Border)	15.96
Krishna	6.64
Mahanadi	4.08
Mahi	7.22
Minor Rivers Draining Into Myanmar and Bangladesh	1.53
Narmada	5.76
Pennar	10.29
Sabarmati	21.67
Subarnarekha	4.17
Tapi	8.80
West Flowing Rivers From Tadri To Kanyakumari	6.27
West Flowing Rivers From Tapi To Tadri	4.42
West Flowing Rivers Of Kutch and Saurashtra Including	13.85

- **Insights:**
  - Basins such as Area of Inland Drainage in Rajasthan and Sabarmati show very high average groundwater depth, while basins like Irrawaddy Drainage and Brahmani–Baitarani show shallow levels.
  - High depth in Rajasthan and western river basins is mainly due to low rainfall, arid climate, and over-extraction, whereas eastern and north-eastern basins benefit from high rainfall and better natural recharge.
  - If current usage patterns continue, inland and western basins are likely to face further groundwater decline, increasing water scarcity in these regions.
  - Water-stressed basins should be prioritized for artificial recharge, rainwater harvesting, and regulated pumping to improve groundwater sustainability.

- **District-wise Percentage of Depth_Category:**

High-Risk Districts

Count of ID	Column Labels		
District_Name	Deep	Medium	Shallow
Alwar	77%	23%	0%
Banas Kantha	64%	33%	3%
Barnala	100%	0%	0%
Churu	94%	6%	0%
Gandhinagar	88%	12%	0%
Jhunjhunu	100%	0%	0%
Mahendragarh	50%	50%	0%
Mahesana	41%	39%	20%
Nagaur	85%	15%	0%
Sikar	95%	4%	1%

Low Risk districts

Count of ID	Column Labels	
District_Name	Medium	Shallow
East Khasi Hills	0%	100%
Jorhat	0%	100%
Karimganj	0%	100%
Kendrapara	2%	98%
Lakhimpur	0%	100%
Nalbari	0%	100%
Reasi	0%	100%
Sivasagar	0%	100%
West Jaintia Hills	0%	100%
West Khasi Hills	0%	100%

- **Insights:**
  - Districts such as Barnala, Jhunjhunu, Churu, Sikar, and Gandhinagar exhibit more than 85% deep-category groundwater levels, while districts in North-Eastern India show nearly 100% shallow groundwater levels.
  - High-risk districts are concentrated in arid and semi-arid regions of Rajasthan, Punjab, and Gujarat, where low rainfall, intensive irrigation, and heavy groundwater extraction dominate, whereas low-risk districts benefit from high rainfall and strong natural recharge.
  - If current groundwater extraction continues, high-risk districts are likely to experience further deepening of water tables and increased risk of water scarcity.
  - Immediate implementation of recharge structures, regulated borewell usage, and promotion of water-efficient agriculture should be prioritized in high-risk districts, while sustainable practices should be maintained in low-risk districts.

- **Seasonal and Year-wise Groundwater Depth Analysis (2019–2021):**

Average of Current_Depth_to_Water_Level	Column Labels			
Monsoon_Season	2019	2020	2021	Grand Total
Post-Monsoon	7.49	8.84	8.90	8.02
Pre-Monsoon	8.52	8.01	8.26	8.30

- **Insights:**
  - Post-monsoon groundwater levels are lower than pre-monsoon levels in all three years, indicating seasonal recharge.
  - The rising post-monsoon depth from 2019 to 2021 suggests reduced rainfall effectiveness and increased extraction.
  - If the current trend continues, post-monsoon groundwater levels are likely to deepen further in coming years.

- **Artificial recharge** and **groundwater regulation** should be strengthened to improve post-monsoon recovery.

- **State-wise Groundwater Stress Identification (2019–2021):**

Average of Current_Depth_to_Water_Level	Column Labels		
State_Name	2019	2020	2021
Andaman and Nicobar Islands	2.95	5.56	6.89
Andhra Pradesh	6.85	4.46	4.03
Arunachal Pradesh	4.37	4.89	4.90
Assam	4.37	4.67	3.59
Bihar	4.84	3.70	4.34
Chandigarh	19.97	30.58	22.20
Chhattisgarh	5.53	4.33	5.95
Dadra and Nagar Haveli and Daman and Diu	4.85	3.56	4.17
Delhi	15.12	14.75	14.37
Goa	4.76	4.97	5.29
Gujarat	13.68	10.42	11.48
Haryana	17.36	24.91	24.94
Himachal Pradesh	5.65	5.29	6.66
Jammu and Kashmir	4.98	4.28	4.62
Jharkhand	5.34	3.96	5.30
Karnataka	7.16	6.19	5.89
Kerala	6.14	6.45	6.83
Madhya Pradesh	7.52	5.84	8.03
Maharashtra	7.07	5.19	5.93
Meghalaya	2.78	2.45	2.75
Nagaland	3.26	4.55	3.59
Odisha	3.23	2.87	3.69
Puducherry	11.41	12.34	10.60
Punjab	15.89	16.81	16.16
Rajasthan	21.87	20.54	20.99
Tamil Nadu	9.89	10.17	9.67
Telangana	10.05	6.15	5.81
Tripura	7.22	7.31	7.74
Uttar Pradesh	10.06	10.58	7.98
Uttarakhand	12.89	11.55	12.56
West Bengal	6.29	5.28	6.51

- **Insights:**
  - Rajasthan, Haryana, Chandigarh, Punjab, Delhi, and Gujarat consistently record the highest average groundwater depth, making them the **worst-performing states**.
  - These states experience severe groundwater depletion due to over-dependence on borewells, water-intensive crops, rapid urbanization, and low recharge capacity.
  - Without corrective measures, these worst-performing states are likely to face increased well failures and acute water scarcity in the near future.
  - Targeted interventions such as **crop diversification**, **regulated groundwater extraction**, and **large-scale recharge projects** should be prioritized in these states.

- **Percentage Distribution of Groundwater Depth Categories by State:**

Count of ID	Column Labels		
Depth Category	Deep	Medium	Shallow
Andaman and Nicobar Islands	4.38%	13.75%	81.88%
Andhra Pradesh	6.03%	26.38%	67.60%
Arunachal Pradesh	0.00%	30.91%	69.09%
Assam	0.43%	30.36%	69.21%
Bihar	0.86%	30.95%	68.19%
Chandigarh	57.81%	21.88%	20.31%
Chhattisgarh	1.98%	39.20%	58.82%
Dadra and Nagar Haveli and Daman and Diu	0.55%	31.15%	68.31%
Delhi	27.93%	50.45%	21.62%
Goa	0.22%	38.77%	61.01%
Gujarat	18.56%	36.60%	44.84%
Haryana	60.65%	20.55%	18.80%
Himachal Pradesh	6.81%	32.34%	60.85%
Jammu and Kashmir	5.76%	20.73%	73.51%
Jharkhand	0.46%	42.20%	57.33%
Karnataka	6.52%	42.90%	50.57%
Kerala	3.66%	51.93%	44.41%
Madhya Pradesh	7.31%	45.77%	46.93%
Maharashtra	6.06%	44.27%	49.66%
Meghalaya	0.00%	4.73%	95.27%
Nagaland	0.00%	18.75%	81.25%
Odisha	0.08%	17.25%	82.67%
Puducherry	40.00%	8.00%	52.00%
Punjab	44.11%	34.96%	20.93%
Rajasthan	43.89%	32.15%	23.96%
Tamil Nadu	12.37%	42.75%	44.88%
Telangana	15.32%	38.40%	46.28%
Tripura	17.33%	16.83%	65.84%
Uttar Pradesh	26.45%	27.73%	45.82%
Uttarakhand	21.59%	40.63%	37.78%
West Bengal	7.62%	34.09%	58.30%

- **Insights:**
  - **North-Eastern states** show over **80% shallow groundwater levels**, while Haryana, Chandigarh, Rajasthan, and Punjab show high deep-category proportions.
  - High deep-category concentration is linked to intensive irrigation, urban water demand, and limited recharge capacity.
  - If current trends continue, high-risk states are likely to experience further expansion of deep groundwater zones.
  - Water conservation, regulated extraction, and artificial recharge should be prioritized in high deep-category states.

- **Percentage Distribution of Groundwater Trend_Type by State:**

Count of ID	Column Labels	
Station_Name	Decine	Recovery
Andaman and Nicobar Islands	32.81%	67.19%
Andhra Pradesh	50.43%	49.57%
Arunachal Pradesh	40.00%	60.00%
Assam	47.15%	52.85%
Bihar	40.71%	59.29%
Chandigarh	53.91%	46.09%
Chhattisgarh	39.13%	60.87%
Dadra and Nagar Haveli and Daman and Diu	34.43%	65.57%
Delhi	55.50%	44.50%
Goa	38.11%	61.89%
Gujarat	42.73%	57.27%
Haryana	47.42%	52.58%
Himachal Pradesh	36.88%	63.12%
Jammu and Kashmir	51.23%	48.77%
Jharkhand	40.96%	59.04%
Karnataka	42.62%	57.38%
Kerala	33.64%	66.36%
Madhya Pradesh	36.93%	63.07%
Maharashtra	40.15%	59.85%
Meghalaya	41.89%	58.11%
Nagaland	37.50%	62.50%
Odisha	34.95%	65.05%
Puducherry	50.00%	50.00%
Punjab	49.95%	50.05%
Rajasthan	39.91%	60.09%
Tamil Nadu	48.56%	51.44%
Telangana	47.45%	52.55%
Tripura	36.14%	63.86%
Uttar Pradesh	43.31%	56.69%
Uttarakhand	43.75%	56.25%
West Bengal	44.01%	55.99%

  - Conditional formatting was applied to the Decline vs Recovery percentage table to enhance visual interpretation.
  - States with decline proportions exceeding 50% were highlighted in red to indicate high groundwater stress.
  - Higher recovery percentages were highlighted in green to represent stable recharge conditions. This visual encoding enables quick identification of high-risk and relatively safe regions.

- **Insights:**
  - Most Indian states show **higher groundwater recovery percentages than decline**, indicating overall seasonal recharge during 2019–2021.
  - States such as Delhi, Chandigarh, Andhra Pradesh, and Jammu & Kashmir exhibit decline percentages above 50%, reflecting localized over-extraction and urban–agricultural pressure.
  - States with decline proportions exceeding or nearing 50% are likely to experience increasing groundwater stress if current usage patterns continue.
  - Focused groundwater regulation, recharge infrastructure, and water-use efficiency measures should be prioritized in high-decline states.

- **Basin-wise Groundwater Decline vs Recovery Percentage Analysis (2019–2021)**

Count of ID	Column Labels	
Basin_Name	Decine	Recovery
Area Of Inland Drainage In Rajasthan	44.85%	55.15%
Barak and Others	40.54%	59.46%
Brahmani and Baitarani	35.23%	64.77%
Brahmaputra	44.92%	55.08%
Cauvery	45.18%	54.82%
Drainage Area Of Andaman and Nicobar Islands Basin	32.81%	67.19%
East Flowing Rivers Between Mahanadi and Pennar	46.48%	53.52%
East Flowing Rivers Between Pennar and Kanyakumari	48.96%	51.04%
Ganga	41.29%	58.71%
Godavari	40.06%	59.94%
Indus (Up To Border)	48.38%	51.62%
Krishna	46.66%	53.34%
Mahanadi	36.59%	63.41%
Mahi	38.32%	61.68%
Minor Rivers Draining Into Myanmar and Bangladesh	36.84%	63.16%
Narmada	39.25%	60.75%
Pennar	52.52%	47.48%
Sabarmati	44.71%	55.29%
Subarnarekha	39.54%	60.46%
Tapi	42.33%	57.67%
West Flowing Rivers From Tadri To Kanyakumari	34.67%	65.33%
West Flowing Rivers From Tapi To Tadri	35.61%	64.39%
West Flowing Rivers Of Kutch and Saurashtra Including Luni	43.81%	56.19%

- Applied color-scale conditional formatting to highlight basin-wise groundwater trend patterns.
- Used red shades to indicate high decline proportions and green shades to represent strong recovery.
- Medium shades were used to represent transitional basins with balanced trends.

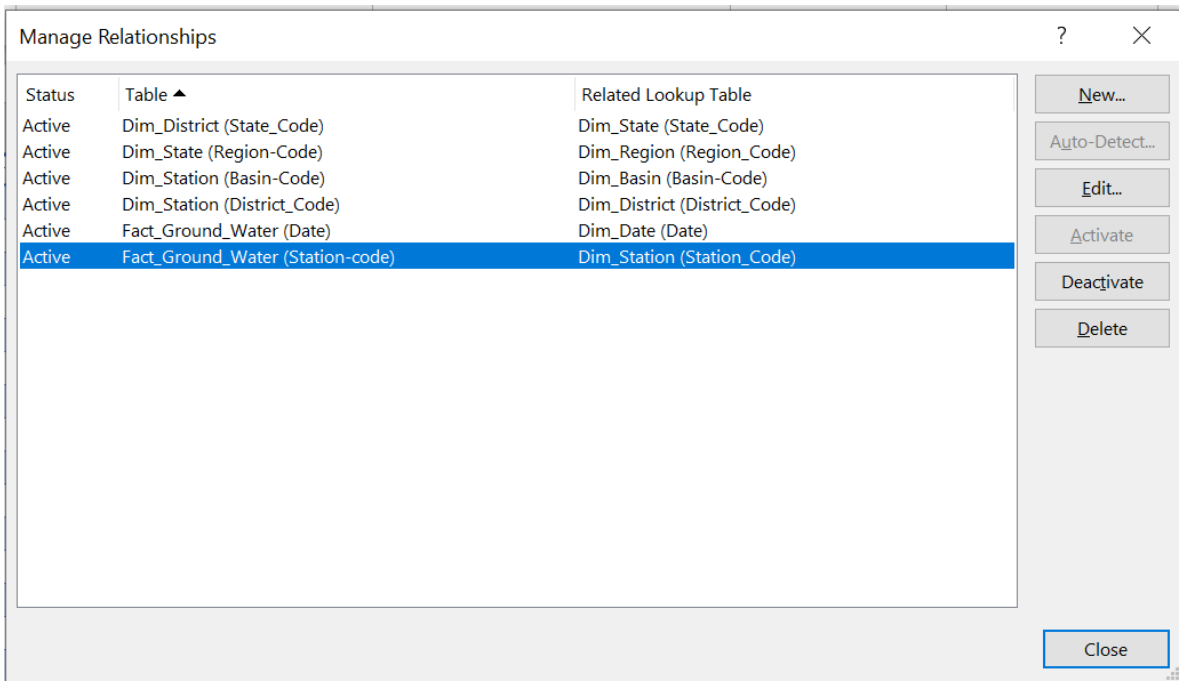
- **Insights:**
  - Most river basins show higher recovery percentages than decline, indicating overall seasonal groundwater recharge.
  - Basins such as Pennar, East Flowing Rivers (Pennar–Kanyakumari), and Indus (Up to Border) exhibit higher decline proportions, reflecting intensive irrigation and extraction pressure.
  - Basins with decline levels approaching or exceeding 50% are likely to experience increasing groundwater stress in future years.
  - Basin-level recharge planning and regulated groundwater abstraction should be prioritized in high-decline river basins.

- **Statistical Analysis (Using Analysis ToolPak):**

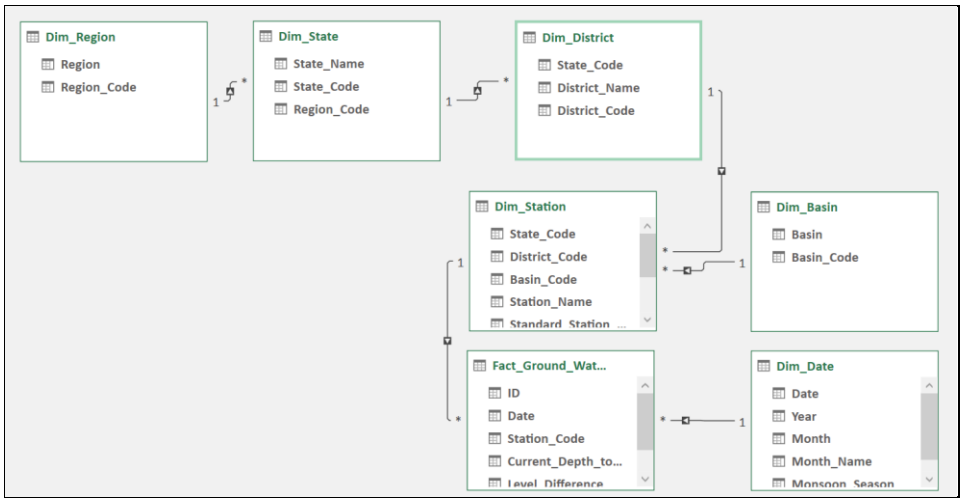
Descriptive Statistics	Current_Depth_to_Water_Level	Interpretation
Mean	8.24	On average, groundwater is found at about 8.24 meters below the ground level.
Standard Error	0.03	The calculated average depth is reliable and accurate.
Median	4.9	In half of the locations, groundwater depth is less than 4.9 meters, and in the other half, it is more than 4.9 meters.
Mode	2.5	The most commonly observed groundwater depth is around 2.5 meters, indicating many shallow water locations.
Standard Deviation	11.18	Groundwater depth varies greatly from place to place, showing uneven water availability across regions.
Kurtosis	42.05	There are some extreme cases where groundwater levels are unusually deep, showing highly stressed zones.
Skewness	4.83	Most areas have shallow groundwater, but a few areas have very deep levels, which increases the average value.
Range	299.3	The difference between the shallowest and deepest groundwater levels is very high, indicating large regional differences.
Minimum	0	In some areas, groundwater is available very close to the surface, showing good water conditions.
Maximum	299.3	In some areas, groundwater is found very deep, showing severe water scarcity.
Count	125312	A large amount of data was used, so the analysis is dependable.

7. Data Modelling and DAX (Power BI)

- Relationships between dimension and fact tables were established using Power Pivot based on primary and foreign key mappings.
- All relationships were defined as **one-to-many (1:*)**, where dimension tables represent the "one" side and the fact table represents the "many" side.
- Administrative and hydrological dimensions are linked to the Fact_Ground_Water table through the Station dimension, while the Date dimension is directly connected to the fact table to support time-based analysis.
- This design ensures data consistency, efficient filtering, and reliable multi-level analysis.
- **Active Relationships Between Fact and Dimension Tables in Power Pivot:** The below screenshot shows the active one-to-many relationships created in Power Pivot, linking fact and dimension tables to support hierarchical and time-based analysis.



- **Hybrid Snowflake Schema – Diagram View of Data Model**



- The above figure presents the Diagram View of the data model created using Power Pivot.
- It illustrates the hybrid snowflake schema adopted in this project, showing the hierarchical relationships among Region, State, District, Station, Basin, Date, and Fact tables.
- The Fact_Ground_Water table is centrally connected to the Dim_Station and Dim_Date tables, while the Station dimension serves as a bridge between administrative and hydrological hierarchies.
- This structure supports multi-level, time-based, and basin-wise analysis while maintaining normalization and data integrity.

- **DAX Measures:** DAX (Data Analysis Expressions) was used to compute key groundwater monitoring and risk assessment metrics. The following measures were created:
  - **Total Monitored Stations:** Calculates the total number of distinct monitoring stations in the selected filter context.

```
Total Monitored Stations = DISTINCTCOUNT(Fact_Ground_Water[Station_Code])
```

  - **Total Deep Stations:** Calculates the number of stations categorized as "Deep".

```
Total Deep Stations = CALCULATE(DISTINCTCOUNT(Fact_Ground_Water[Station_Code]),FILTER(Fact_Ground_Water, Fact_Ground_Water
[Depth_Category]="Deep"))
```

  - **Percentage of Deep Stations:** Computes the proportion of deep stations relative to total monitored stations.

```
% Deep Stations = IF(
    [Data Coverage %] < 0.1,
    BLANK(),
    DIVIDE([Total Deep Stations], [Total Monitored Stations])
)
```

  - **Data Coverage %:** Calculates the percentage of monitored stations that have available groundwater data within the selected filter context.

```
Data Coverage % =
DIVIDE(
    [Total Monitored Stations],
    DISTINCTCOUNT(Dim_Station[Station_Code])
)
```

This measure ensures that groundwater analysis is performed only when sufficient data is available, improving reliability and preventing misleading conclusions.

- **Average Current Depth to Water Level:** Computes the average groundwater depth for the selected filters, only when sufficient data coverage exists.

```
Avg Current_Depth_to_Water_Level = IF(
    [Data Coverage %] < 0.1,
    BLANK(),
    AVERAGE(Fact_Ground_Water[Current_Depth_to_Water_Level])
)
```

- **Risk Category Classification:** Classifies regions based on percentage of deep stations.

```
Risk Category = SWITCH(
    TRUE(),
    [Data Coverage %] < 0.1, "Insufficient Data",
    [% Deep Stations] < 0.2,"Safe",
    [% Deep Stations] < 0.4,"Moderate",
    [% Deep Stations] < 0.6,"Critical",
    "Overexploited")

```

- **Calculated Columns for Location Hierarchy:** To enable drill-down analysis across administrative levels, calculated columns were created within the Dim_Station table. Since Power BI hierarchies require all levels to exist within the same table, location attributes were brought into Dim_Station using the RELATED() function.
  - **Country:** Creates a constant top-level attribute for the location hierarchy.

```
Country = "India"
```

    The dataset represents groundwater monitoring stations across India only. Adding a Country column enables structured drill-down starting from the national level.
  - **Region:** Fetches the Region value from the Dim_Region table into Dim_Station.

```
Region = RELATED(Dim_Region[Region])
```

The RELATED() function retrieves data from a related dimension table based on existing relationships in the data model.

  - **State Name:** Brings the State attribute into Dim_Station for hierarchical grouping.

```
State_Name = RELATED(Dim_State[State_Name])
```

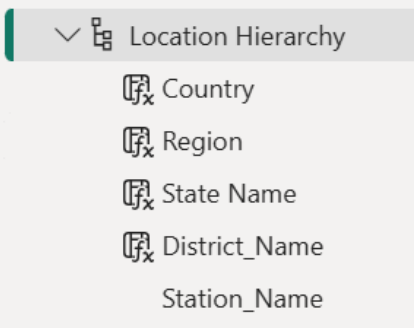
Create a new table by typing or pasting in new co

  - **District Name:** Brings the District attribute into the same table to complete the administrative hierarchy.

```
District_Name = RELATED(Dim_District[District_Name])
```

- **Location Hierarchy**
  - A Location Hierarchy was created to enable structured drill-down analysis across administrative levels.
  - The hierarchy includes: Country → Region → State → District → Station_Name.

- It allows multi-level analysis from national trends to individual monitoring stations.
- The hierarchy enhances interactive exploration and supports both aggregated and granular groundwater analysis.
- It enables identification of regional stress zones as well as localized station-level variations.



- **Calculated Column Trend Type Classification:**
  - A calculated column Trend_Type was created to classify groundwater level changes.
  - Classification is based on Level_Difference (Current Year – Previous Year).
  - Stations are categorized as Stable, Decline, or Recovery.
  - This enables comparison of depletion and recharge patterns across regions and states.
  - The column supports trend-based visuals such as depletion vs recovery analysis.

```
Trend_Type = IF(Fact_Ground_Water[Level_Difference]=0,"Stable",IF(Fact_Ground_Water[Level_Difference]>0,"Decline","Recovery"))
```

8. Analysis and Visualizations with Insights (Power BI)

- **Dashboard Features:**
  - **Multiple Visualizations Based on Problem Statement**

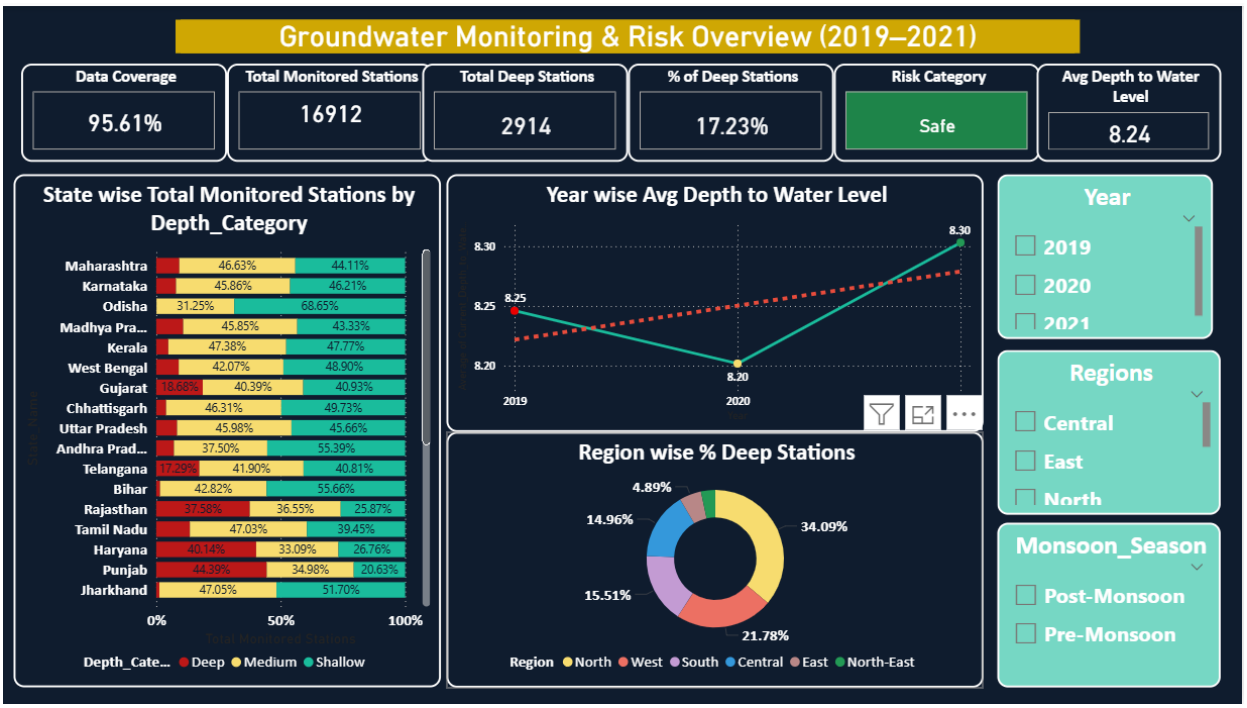
    The dashboard includes multiple visual elements designed to analyze groundwater trends and risk levels:
    - **KPI Cards –** Display key metrics such as Data Coverage %, Total Monitored Stations, Total Deep Stations, % Deep Stations, Risk Category, and Average Depth.
    - **100 % Stacked Bar Charts –** Show depth category distribution (Deep, Medium, Shallow) across states.
    - **Line Chart –** Represents year-wise average groundwater depth trends.
    - **Donut Chart –** Displays region-wise percentage of deep stations.
    - **Map Visual –** Shows spatial distribution of groundwater stations.
    - **Matrix Visual –** Presents risk classification across administrative hierarchy.

- The Risk Classification matrix supports hierarchical drill-down from national to district level.
- Station-level drill-down was excluded, as risk category is derived from aggregated % of Deep stations.
- Risk evaluation is therefore performed at administrative levels where percentage-based classification is meaningful.

- **Stacked Column Chart (Groundwater Depletion vs Recharge by State & Basin-wise Depth Distribution)** – One chart Compares Decline and Recovery trends at the state level and another column chart analyses groundwater depth variation across major river basins

- These visuals were selected to communicate trends, comparisons, proportions, and spatial patterns effectively.
- **Interactive Report Layout:** The dashboard was designed with interactive features including:
  - Drill-down functionality using Location Hierarchy (Country → Region → State → District → Station).
  - Slicers for Year, Region, and Season Type.
  - Cross-filtering interaction between visuals.
  - Dynamic KPI updates based on selected filters.
  - This enables flexible multi-level groundwater analysis.
- **Consolidated Report/Dashboard:**

Groundwater Monitoring & Risk Overview (2019-2021)



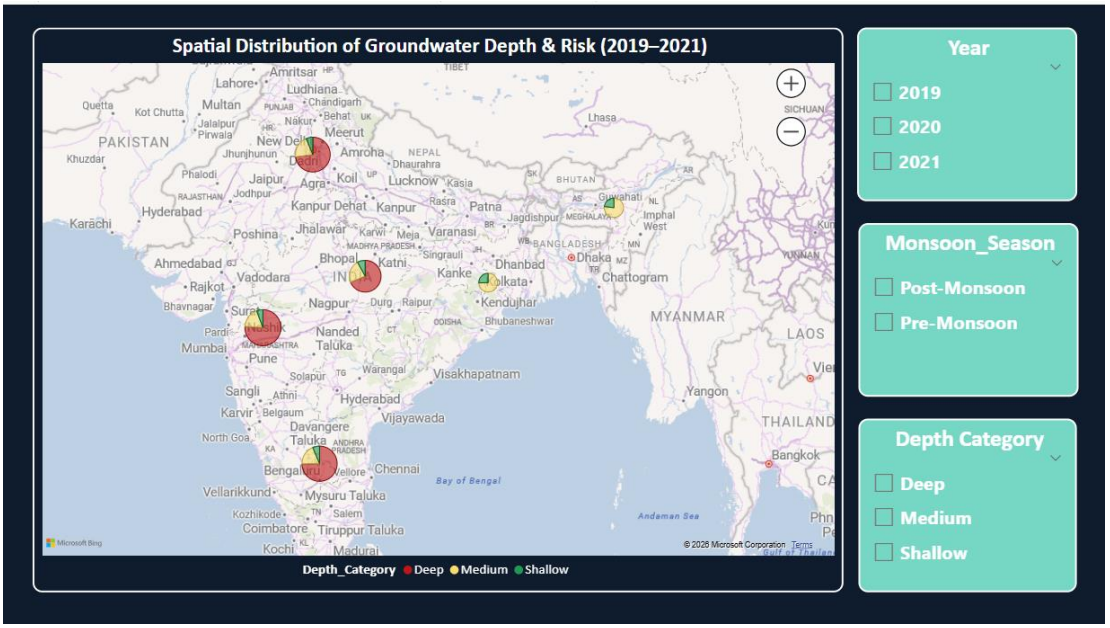
Key Findings:

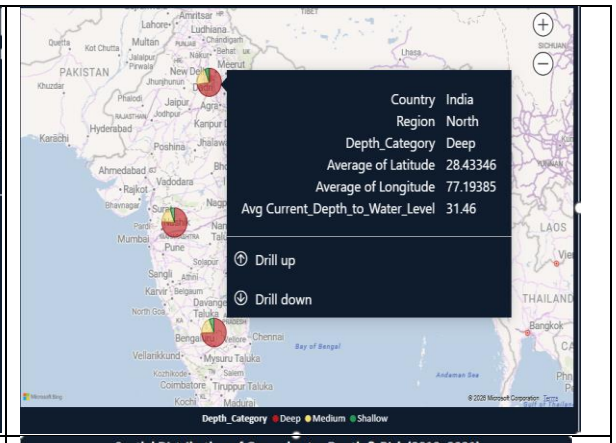
- Overall data coverage is high (above 95%), ensuring reliable groundwater monitoring analysis.
- Approximately 17% of monitored stations fall under the Deep category, indicating moderate groundwater stress at the national level.
- Groundwater depth shows a slight increasing trend from 2019 to 2021, with a dip observed in 2020.
- Certain states show a higher proportion of Deep category stations compared to others, indicating uneven groundwater stress distribution.
- Regional variation is evident, with some regions contributing more significantly to deep groundwater levels.

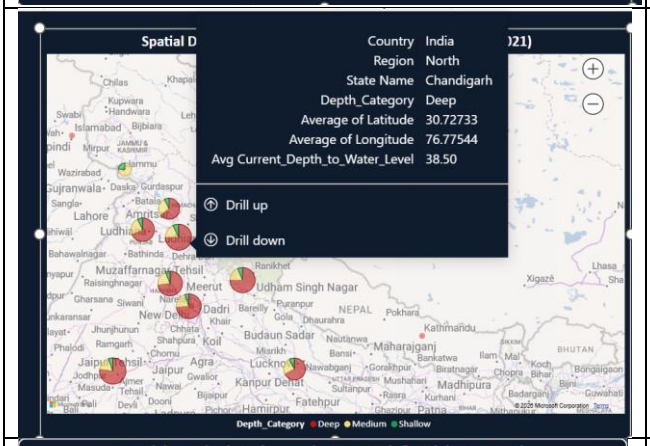
Insights:

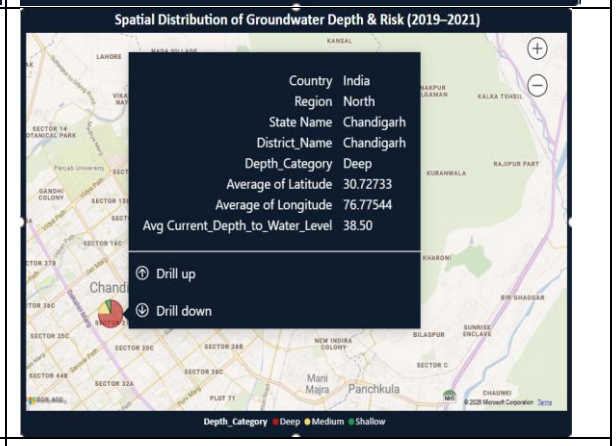
- Groundwater depth and stress levels vary across regions and states.
- A measurable proportion of stations fall under the Deep category.
- The overall trend from 2019–2021 shows minor fluctuations with slight recovery.
- Regional differences indicate uneven recharge and usage patterns.
- States with higher proportions of deep stations reflect localized groundwater stress conditions.
- Regional variations highlight differences in recharge capacity and extraction intensity.
- Continued imbalance between extraction and recharge may increase groundwater risk in moderately stressed regions.
- Targeted recharge initiatives and regulated groundwater usage are essential to maintain sustainability.

Spatial Distribution of Groundwater Depth & Risk (2019-2021)









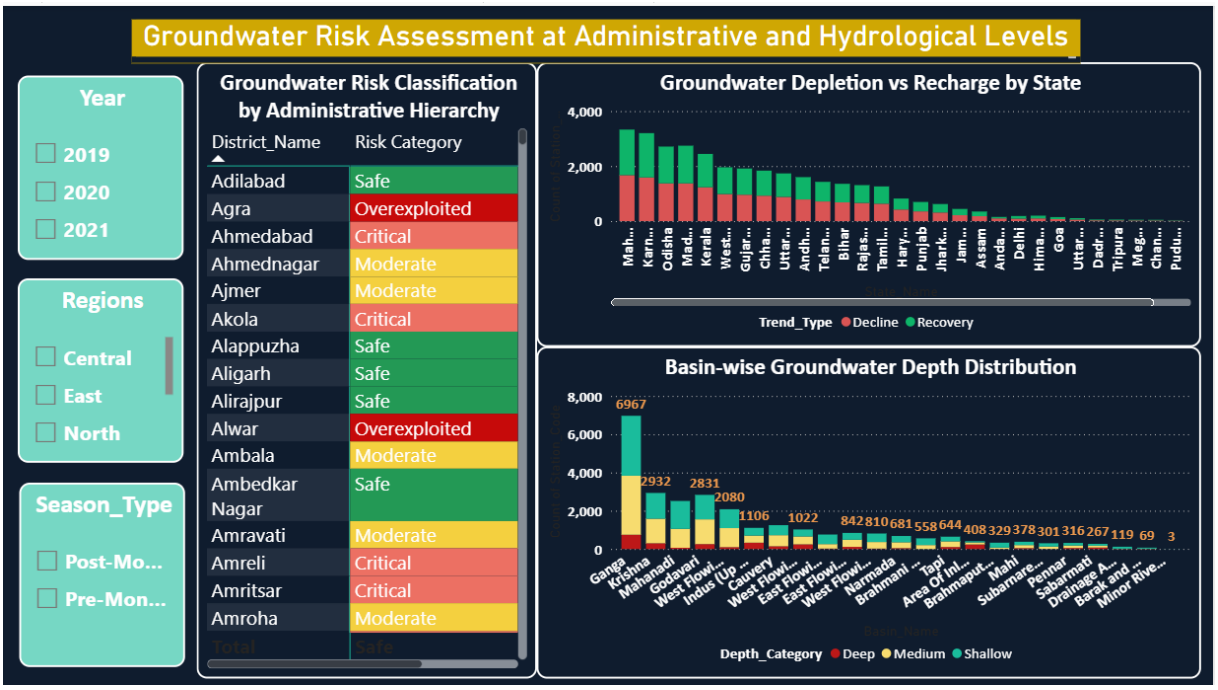
- All these visuals from country→Region→State→District→Station attained using drill down.

- **Key Findings**
  - Groundwater depth distribution varies significantly across regions of India.
  - Certain clusters show higher concentration of Deep category stations.
  - Spatial patterns indicate localized groundwater stress rather than uniform distribution.
  - Drill-down capability allows analysis from country level down to individual station level.

- **Insights**
  - Groundwater stress is spatially uneven across India.
  - Some regions display higher proportions of Deep stations.
  - Interactive drill-down enables hierarchical analysis (Country → Region → State → District → Station).
  - Spatial clustering of deep stations indicates concentrated groundwater stress zones.

- Areas with moderate stress require early intervention to prevent further depletion.
- Drill-down capability enables identification of priority districts for targeted groundwater management.
- Region-specific recharge and monitoring strategies are necessary in high-stress clusters.

Groundwater Risk Assessment at Administrative and Hydrological Levels:



- **Key Findings**
  - Groundwater risk levels vary significantly across districts.
  - Several districts fall under Critical and Overexploited categories.
  - Depletion trends are higher than recovery in multiple states.
  - Basin-wise analysis shows uneven groundwater depth distribution across river basins.
- **Insights**
  - Risk categories range from Safe to Overexploited across districts.
  - Certain states show higher counts of declining stations.
  - Major river basins display varying proportions of Deep, Medium, and Shallow stations.
  - Districts classified as Critical and Overexploited indicate sustained extraction pressure.
  - Basins with higher deep-station concentration reflect hydrological imbalance.
  - Immediate regulatory and recharge measures are required in high-risk districts and stressed basins.
  - Basin-level planning should complement administrative groundwater management for long-term sustainability.

9. Final Conclusion:

- Groundwater levels show significant variation across India.
- North and Western states such as Rajasthan, Haryana, Punjab, and Delhi face higher groundwater stress.
- North-Eastern and Eastern states show relatively healthier groundwater conditions due to better natural recharge.
- District-level analysis identified severe stress zones in arid and semi-arid regions.
- Seasonal analysis indicates that post-monsoon recharge exists, but its effectiveness is gradually reducing.
- Statistical results confirm high variability and the presence of extreme groundwater depletion areas.
- Sustainable groundwater management and regulated extraction are necessary to prevent future water scarcity.
- The project was executed using Microsoft Excel for data cleaning, transformation, preprocessing, and schema preparation, and Power BI for data modelling, DAX calculations, visualization, and interactive dashboard development.
- Overall, the analysis demonstrates how data analytics tools can be effectively applied to support sustainable groundwater management and informed decision-making.